

Table 1: End-to-End Pipeline Latency ($N = 50$)

Phase	Mean	σ	Median
Authentication (JWT + DB)	84.17 ms	7.26 ms	82.50 ms
TOTP setup (QR generation)	40.68 ms	1.28 ms	40.44 ms
TOTP step-up verification	51.86 ms	6.92 ms	49.95 ms
Job submission (SFTP + bsub)	5220.97 ms	833.45 ms	4840.31 ms
LSF completion poll	2014.12 ms	1405.24 ms	1110.21 ms
Total pipeline	7434.46 ms	1570.88 ms	6654.13 ms

Table 2: Security Layer Component Latency ($N = 100$)

Component	Mean	σ	Median
JWT verification (Keycloak RS256)	38.70 ms	1.52 ms	38.45 ms
AST scan — simple script	59.92 μ s	12.37 μ s	57.50 μ s
AST scan — medium script	243.68 μ s	27.66 μ s	236.35 μ s
AST scan — complex script	340.49 μ s	29.21 μ s	330.90 μ s
AST scan — malicious (early exit)	26.49 μ s	3.55 μ s	25.70 μ s
Sandbox wrapper generation	0.27 μ s	—	0.20 μ s
LSF script generation	2.73 μ s	—	2.50 μ s
Total security overhead	≈ 39 ms		

Table 3: CARTA Risk Engine Scenario Verification

S#	Scenario	Signals	ρ_{req}	ρ_{sess}	Action	P
S1	Legitimate baseline	—	0.000	0.000	allow	
S2	Credential attack	R7=1.0	0.350	0.105	mfa	
S3	Rate flood	R5+combo $\times$ 1.25	0.350	0.105	mfa	
S4	Velocity pattern	R5=0.7	0.350	0.105	mfa	
S5	Multi-signal combo	R5+R7 $\times$ 1.45	0.441	0.132	mfa	
S6	High rate	R5=1.0	0.441	0.132	mfa	
S7	Session EMA decay	0.441 $\rightarrow$ 0.020	—	—	allow	
S8	Tamper detection	280 entries, 28	—	—	detected	
FPR = 0/1 = 0.0% TPR = 6/6 = 100% Tamper = 1/1 = 100%						